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B.Tech. AI/DS Mid-Term Examinations – October 2025

First Semester

Artificial Intelligence & Data Science

23PHY104 Computational Mechanics-I – Batch 1

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Apply numerical methods to develop computational models for mechanical systems and analyze their behavior
CO02	Derive constitutive relations for mechanical systems in motion or at rest, including particles and rigid bodies, and use these equations to solve real world problems.
CO03	Evaluate the results of computational simulations and use this information to make informed decisions about mechanical systems design and optimization
CO04	Use software tools for computational mechanics, including code for solving equations of motion and simulating mechanical systems

Answer all questions

Part A

1) What is Newton's second law of motion? Use this to discuss why an astronaut inside a spacecraft orbiting Earth experiences weightlessness even though the Earth's gravitational force acts on them.

[2] [CO02][BTL 2]

2) What are two fundamental properties of any general rotation matrix?

[2] [CO01][BTL 1]

3) Find a unit vector perpendicular to both $\vec{A} = \hat{\imath} + \hat{\jmath} - \hat{k}$ and $\vec{B} = 2\hat{\imath} - \hat{\jmath} + 3\hat{k}$.

[2] [CO01][BTL 3]

4) Explain why a mass located farther from the pivot makes a rotating body harder to accelerate angularly, even if the total mass remains the same. Use this reasoning to justify why the quantity $I = \int r^2 dm$ is called the moment of 'inertia'.

[2] [CO02][BTL 4]

5) Define Euler integration and Verlet integration. Is there any advantage of using the Verlet integration over Euler integration for computational mechanics?

[2] [CO03][BTL 1]

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Part B

6) What is a quaternion? If we have quaternions $q_1 = 2 + 3i - j + k, q_2 = i + 4j$, what is the value of $q_1 q_2$? For any two quaternions, when is $\mathrm{Im}(q_1 q_2) = \mathrm{Im}(q_2 q_1)$?

[5] [CO02][BTL 1]

7) A rod of length L has a non-uniform density λ, the mass per unit length of the rod, which varies as $\lambda = \frac{\lambda_0 s}{L}$, where $\lambda_0 > 1$ is a constant and s is the distance from the right-end of the rod. Find the centre-of-mass.

[5] [CO02][BTL 5]

8) Calculate the line integral for $\vec{F} = xy^2\hat{x} + (x - yx)\hat{y}$ along the curve $y = 2x^2$ from $r_1 = [-3,9]$ to $r_2 = [2,4]$.

[5] [CO01][BTL 3]

9) State and prove the Euler's theorem for general rotations.

[5] [CO02][BTL 4]

Part C

10) Define the acceleration of a particle in three-dimensional space and explain how it can be decomposed into tangential and normal components.

[2] [CO02][BTL 2]

Derive the expression for the radius of curvature of a particle moving in three-dimensional space, at any general point in time.

[4] [CO02][BTL 4]

A particle moves along a helix defined by $\vec{r}(t) = a\cos(\omega t)\,\hat{\imath} + a\sin(\omega t)\,\hat{\jmath} + bt\hat{k}$. Compute the tangential and normal components of acceleration at time t.

[4] [CO02][BTL 3]

11) Define curl, gradient and divergence. Evaluate $\vec{\nabla}.\vec{A}\hat{\imath} + \vec{\nabla} \times \vec{A} + \vec{\nabla}(\vec{\nabla}.\vec{A})$ for $\vec{A} = (2x - z)\hat{\imath} + xy^2 z\hat{\jmath}$ at (1,1,1).

[5] [CO01][BTL 5]

Evaluate $\vec{\nabla} \times \vec{\nabla}\phi$ for some scalar function ϕ. What does this imply for conservative forces?

[5] [CO01][BTL 6]

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